

B. Bitwise Reversion

time limit per test: 1.5 seconds
 memory limit per test: 256 megabytes

You are given three non-negative integers x , y and z . Determine whether there exist three non-negative integers a , b and c satisfying the following three conditions:

- $a \& b = x$
- $b \& c = y$
- $a \& c = z$

where $\&$ denotes the bitwise AND operation.

Input

Each test contains multiple test cases. The first line contains the number of test cases t ($1 \leq t \leq 10^4$). The description of the test cases follows.

The first and only line of each test case contains three integers x , y and z ($0 \leq x, y, z \leq 10^9$) — the target values of $a \& b$, $b \& c$ and $a \& c$, respectively.

Output

For each test case, output "YES" if there exists three non-negative integers a , b , and c satisfying the above conditions, and "NO" otherwise.

You can output the answer in any case (upper or lower). For example, the strings "yEs", "yes", "Yes", and "YES" will be recognized as positive responses.

Example

input	Copy
5 1 1 1 3 2 6 4 8 12 9 10 12 12730 3088 28130	
output	Copy
YES YES NO YES NO	

Note

In the first test case, $a = 3$, $b = 5$, and $c = 9$ satisfies the condition as $3 \& 5 = 1$, $5 \& 9 = 1$, and $3 \& 9 = 1$.

In the second test case, $a = 7$, $b = 3$, and $c = 22$ satisfies the condition as $7 \& 3 = 3$, $3 \& 22 = 2$, and $7 \& 22 = 6$.

In the third test case, it can be proven that there are no three non-negative integers a , b , and c such that $a \& b = 4$, $b \& c = 8$, and $a \& c = 12$.

Codeforces Round 1057 (Div. 2)

Contest is running

01:29:37

Contestant



→ Submit?

Language: Python 3.13.2
 Almost always, if you send a solution on PyPy, it works much faster

Choose file: Choose File No file chosen

Be careful: there is 50 points penalty for submission which fails the pretests or resubmission (except failure on the first test, denial of judgement or similar verdicts). "Passed pretests" submission verdict doesn't guarantee that the solution is absolutely correct and it will pass system tests.

Submit

→ Score table

	Score
Problem A	440
Problem B	660
Problem C	1100
Problem D	1540
Problem E	1980
Problem F	2860
Successful hack	100
Unsuccessful hack	-50
Unsuccessful submission	-50
Resubmission	-50

* If you solve problem on 00:30 from the first attempt

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